HEALTHCONNECT CLINIC
WEEK 4 INITIAL DATA ANALYSIS

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

1. Dataset Overview

The HealthConnect appointment dataset contains **5,000 appointment records and 18 variables**. Each record represents an appointment and contains information relating to patient characteristics, appointment details, booking information, previous appointment history, previous no-shows, reminder activity, distance to the clinic, waiting time and appointment outcome.

The dataset contains **1,696 unique patient IDs**, indicating that some patients have multiple appointment records. This is appropriate for the project because the data is appointment-level rather than limited to one record per patient.

The main outcome variable, appointment_outcome, contains three categories:

- **Attended:** 2,314 appointments
- **No-Show:** 2,423 appointments
- **Cancelled:** 263 appointments

The dataset covers appointment dates from **1 January 2025 to 30 June 2026**.

The **HealthConnect Data Dictionary** was reviewed alongside the dataset to understand the purpose and meaning of the variables. The dataset and supporting project resources provide an appropriate foundation for investigating patterns associated with appointment attendance and missed appointments.

The main analytical objective is to understand the factors associated with no-shows and identify information that could support better appointment management and patient engagement.

2. Data Quality Assessment

An initial review of the dataset was conducted to assess completeness, consistency, validity, uniqueness and readiness for further analysis.

2.1 Missing Values

Most variables in the dataset are complete. Missing values were identified in three fields:

- reminder_channel — **1,366 missing values**
- distance_to_clinic_km — **90 missing values**
- waiting_time_minutes — **60 missing values**

The missing reminder_channel values appear to be intentional rather than necessarily representing poor-quality data. They correspond to records where reminder_sent is recorded as **No**. These records can therefore be represented as **"None"** during cleaning to clearly indicate that no reminder channel was used.

The 90 missing distance values and 60 missing waiting-time values should remain blank/null rather than being replaced with assumed values. This prevents artificial information from being introduced into the analysis.

2.2 Date Formatting and Data Types

The appointment_date field is consistently interpretable as a date and covers the period from January 2025 to June 2026.

The booking_date field requires additional attention. The Data Dictionary specifies an ISO date format, while some values in the dataset are displayed in a different format. This creates a risk of incorrect interpretation depending on regional date settings.

The booking_date field should therefore be standardised and validated before being used for date calculations.

The dataset also contains numerical fields such as booking lead days, previous appointments, previous no-shows, distance and waiting time. These should be retained as numerical data types to support calculations and comparisons.

2.3 Consistency

The main categorical variables use consistent labels.

For example, appointment_outcome is categorised as:

- Attended
- No-Show
- Cancelled

Similarly, reminder status uses Yes/No values, while reminder channels include SMS, WhatsApp and Email.

The main consistency issue identified during the initial review is therefore related to date formatting rather than widespread inconsistent categorical values.

2.4 Duplicate Records

No exact duplicate rows were identified.
The appointment_id field contains **5,000 unique IDs**, matching the total number of appointment records. This indicates that each appointment has a unique identifier.
Repeated patient_id values should not be treated as duplicates because the same patient can legitimately have multiple appointments.

2.5 Validity and Unusual Values

The numerical fields were reviewed for obviously invalid or unreasonable values.
The observed ranges include:

- **Age:** 18–80 years
- **Booking lead time:** 0–60 days
- **Previous appointments:** 0–11
- **Previous no-shows:** 0–5
- **Distance to clinic:** 0.5–45.0 km
- **Waiting time:** 2–68 minutes

No negative values were identified in these numerical fields.
The previous no-show values are also within the expected historical appointment range.

2.6 Overall Data Quality Assessment

Overall, the dataset is suitable for initial exploratory analysis.
The main areas requiring attention before further analysis are:

1. Standardising and validating booking_date.
2. Retaining missing distance and waiting-time values as null rather than creating assumed values.
3. Converting missing reminder_channel values to None where reminder_sent is recorded as No.
4. Documenting these changes in the cleaned dataset.

The original dataset will be preserved, while any cleaned or transformed version will be saved separately, as required by the project instructions.

3. Relevant Variables

The following variables were identified as particularly relevant to investigating appointment attendance and no-show behaviour.

Variable	Relevance
appointment_outcome	Main outcome used to identify attended, no-show and cancelled appointments.
appointment_date	Allows attendance patterns to be examined over time.
booking_date	Can be used to examine booking behaviour once the date format is validated.
booking_lead_days	Measures how far in advance an appointment was booked.
previous_appointments	Provides information about a patient's previous appointment history.
previous_no_shows	Helps investigate whether previous missed appointments are associated with future no-shows.
reminder_sent	Allows comparison of appointments with and without recorded reminders.
reminder_channel	Allows comparison of different reminder channels.
appointment_type	Allows attendance patterns to be compared across appointment types.
appointment_day	Allows attendance patterns to be compared across days of the week.
appointment_time	Allows attendance to be examined across different appointment times.
distance_to_clinic_km	Can be investigated as a possible factor associated with attendance.
waiting_time_minutes	Can be investigated as a possible factor associated with appointment outcomes.
age_group	Allows attendance patterns to be compared across age groups.
gender	Allows demographic differences in attendance patterns to be explored.

4. Business Questions

The following questions will guide the subsequent analysis:

4.1 What factors are associated with patients missing scheduled appointments?

This is the main business question and will guide the investigation of patient, appointment and booking characteristics.

4.2 Is booking lead time associated with appointment attendance or no-show behaviour?

The analysis will examine whether appointments booked further in advance show different attendance patterns from appointments booked closer to the appointment date.

4.3 Are patients with previous no-shows more likely to miss subsequent appointments?

This will investigate whether previous appointment behaviour is associated with future attendance outcomes.

4.4 Is recorded reminder activity associated with appointment attendance?

Attendance outcomes will be compared between appointments with and without recorded reminders.

4.5 Do appointment timing, appointment type, distance, waiting time or demographic characteristics show different attendance patterns?

This will help identify operational or patient characteristics that may be associated with higher or lower no-show rates.

5. Proposed KPIs

The following five KPIs are proposed for subsequent stages of the project. At Week 4, these KPIs are being **identified and justified rather than calculated or visualised**, which is consistent with the assignment requirements.

KPI 1 - No-Show Rate

Definition:
The percentage of scheduled appointments recorded as No-Show.
Business Question:
What proportion of scheduled appointments are being missed?
Why it matters:
This provides a clear measure of the scale of the clinic's missed-appointment problem and can serve as a baseline for future improvement.

KPI 2 - Appointment Attendance Rate

Definition:
The percentage of scheduled appointments recorded as Attended.
Business Question:
What proportion of scheduled appointments are successfully attended?
Why it matters:
This provides a direct measure of appointment attendance and complements the no-show rate.

KPI 3 - Repeat No-Show Rate

Definition:
The proportion of appointments resulting in a No-Show among patients with a recorded history of previous no-shows.
Business Question:
Are patients with previous no-shows more likely to miss future appointments?
Why it matters:
This can help identify whether previous attendance behaviour is an important indicator of future no-show behaviour.

KPI 4 - Reminder-Associated Attendance Rate

Definition:
The attendance rate for appointments with a recorded reminder compared with appointments without a recorded reminder.
Business Question:
Is reminder activity associated with appointment attendance?
Why it matters:
This could help determine whether reminder activity is associated with improved attendance and inform future patient-engagement strategies.

KPI 5 - Average Booking Lead Time

Definition:
The average number of days between appointment booking and the scheduled appointment.
Business Question:
Is booking lead time associated with appointment attendance?
Why it matters:
Understanding booking lead time may reveal scheduling patterns that are associated with missed appointments.

6. Initial Analysis Approach

The proposed analysis will be carried out in stages, beginning with data preparation and progressing towards exploratory analysis and KPI development.

6.1 Data Preparation

Tools: Excel and Power Query

The first stage will involve:

- validating the dataset structure;
- standardising date fields;
- investigating the booking_date formatting issue;
- reviewing missing values;
- checking for duplicate records;
- validating numerical fields;
- standardising relevant categorical values; and
- documenting significant transformations.

The cleaned dataset will be saved separately from the original file.

6.2 Exploratory Data Analysis

Tool: Power BI

Power BI will be used to explore relationships between appointment outcomes and relevant variables.
The analysis will focus on:

- attendance versus no-show patterns;
- booking lead time;
- previous no-shows;
- reminder activity;
- reminder channels;
- appointment type;
- appointment day and time;
- age group;
- gender;
- distance to clinic; and
- waiting time.

The objective will be to identify meaningful patterns that can support the business questions rather than simply produce visualisations.

6.3 KPI Development

Tool: Power BI / DAX

The proposed KPIs will be calculated after the data-quality issues have been addressed.
Where appropriate, DAX measures will be developed to calculate attendance rates, no-show rates and other relevant metrics.

6.4 Segmentation and Comparison

Appointment outcomes will be compared across relevant patient and appointment groups.
This will help determine whether particular groups or appointment characteristics show noticeably different attendance patterns.

6.5 Business Interpretation

The findings will ultimately be interpreted in relation to HealthConnect's business problem.
The aim is to identify evidence that can support improved appointment management, patient engagement and decision-making.

7. Assumptions

The following assumptions will guide the analysis:

1. The dataset represents fictional and anonymized HealthConnect appointment records.
2. The HealthConnect Data Dictionary provides the authoritative definitions of the dataset variables.
3. appointment_outcome accurately represents the recorded outcome of each appointment.
4. reminder_sent represents whether a reminder was recorded as sent; it does not confirm that the patient received, read or acted on the reminder.
5. Missing reminder_channel values associated with reminder_sent = No are treated as indicating that no reminder channel was used.
6. booking_lead_days represents the number of days between booking and the scheduled appointment and will be validated before being relied upon for further analysis.

7. The dataset provides sufficient information for an initial investigation, although it may not capture every factor affecting patient attendance.

8. Limitations

The following limitations may affect the analysis:

- **90 distance values** and **60 waiting-time values** are missing.
- The booking_date field requires validation and standardisation before being used for date-based calculations.
- The dataset may not capture every reason why a patient misses an appointment.
- Reminder records do not confirm whether patients actually received or acted on reminders.
- Some demographic or appointment groups may contain fewer observations than others.
- Relationships identified through the analysis will represent associations and should not automatically be interpreted as causal relationships.

These limitations will be considered when interpreting future findings.

9. Risks and Dependencies

Risks

Data-quality risk:
Incorrect date interpretation or missing values could affect calculations and lead to misleading results.
Interpretation risk:
Observed relationships may be incorrectly interpreted as causal relationships.
Missing-information risk:
Factors influencing appointment attendance may exist outside the variables available in the dataset.
Group-comparison risk:
Unequal group sizes may affect the reliability of comparisons.
Dependencies

The analysis depends on:

- the quality of the HealthConnect appointment dataset;
- accurate interpretation of the Data Dictionary;
- the HealthConnect Clinic Knowledge Base for relevant operational context;
- successful validation and preparation of the data;
- availability of the required fields for KPI development; and
- appropriate use of Excel, Power Query and Power BI.